

EEGvis v2

A browser workbench over the whole MNE-Python pipeline — load any recording, clean it, analyse it, localise it, export a script that reproduces every step.

The whole deck is the v2 proposal. Solid boxes = structure or logic carried from v1; per-note tags flag each as NOW, PROPOSED, or OPEN Q.



V1 → V2, IN ONE LINE

A working prototype hit a ceiling: fixed grid, preprocessing-only toolbar, per-op routes, clipped text. v2 keeps the good substrate (canvas, ledger, theme) and rebuilds the shell around MNE's object graph.

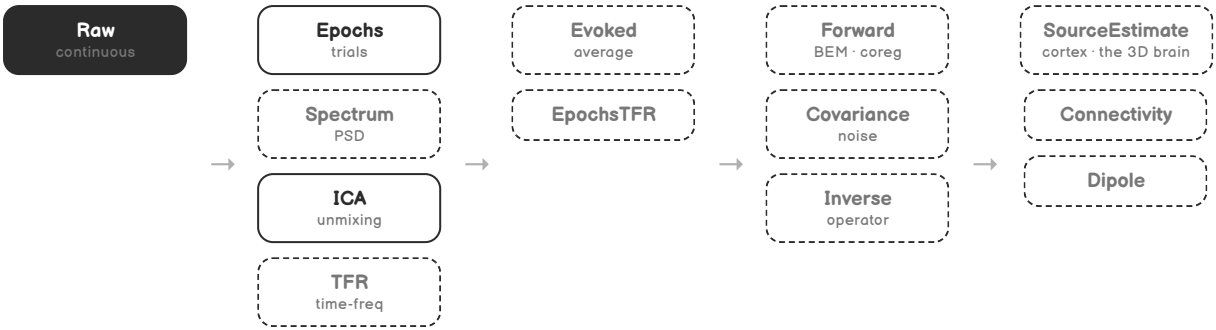
THE ONE BET

One registry entry per MNE operation, one card per MNE plot, one renderer, one provenance graph. Every new capability is *data*, not code. See `BUILD_PLAN_V2.md`.

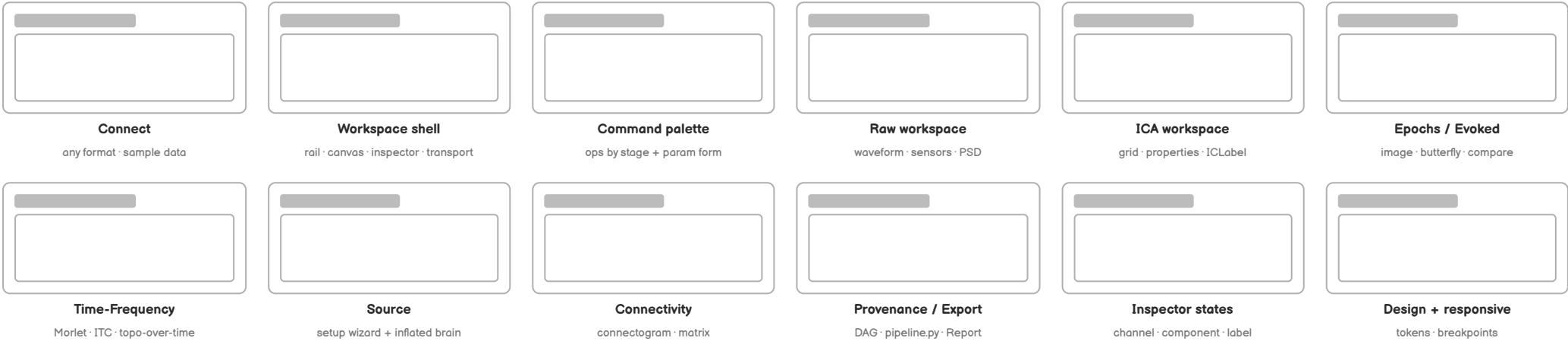
WHAT EVERY SCREEN EARNS

- 1 Container-centric**
The left rail is MNE's object graph: Raw → Epochs → Evoked → Spectrum/TFR → Source. Pick an object, the workspace reshapes to it.
- 2 Dense but not cluttered**
Every label owns its box. Tabular numerals, real breakpoints, one type scale. Density from information, not from cramming.
- 3 Modular canvas**
Views are cards you add / swap / arrange. A workspace preset is just a saved layout. Nothing is hard-wired into the shell.

MNE DATA CONTAINERS — THE LEFT RAIL RENDERS THIS GRAPH; SELECTING A NODE RESHAPES THE WORKSPACE



SCREENS IN THIS DECK



CROSS-SCREEN FLOWS

- **Connect** → **Shell**. Post a file or pick a dataset → routed /s/:id, rail shows Raw.
- **Any op** → **Provenance**. Every registry op appends a DAG node; branch to try alternatives.
- ↔ **Rail node** → **workspace**. Selecting a container swaps the card layout + the ops the command bar offers.
- ↗ **Source rail node** → **wizard**. Gated: montage → forward → covariance → inverse, then the brain card unlocks.

PERSISTENT CHROME: COMMAND BAR (TOP) · CONTAINER RAIL (LEFT) · INSPECTOR (RIGHT) · TRANSPORT + PROVENANCE (BOTTOM)



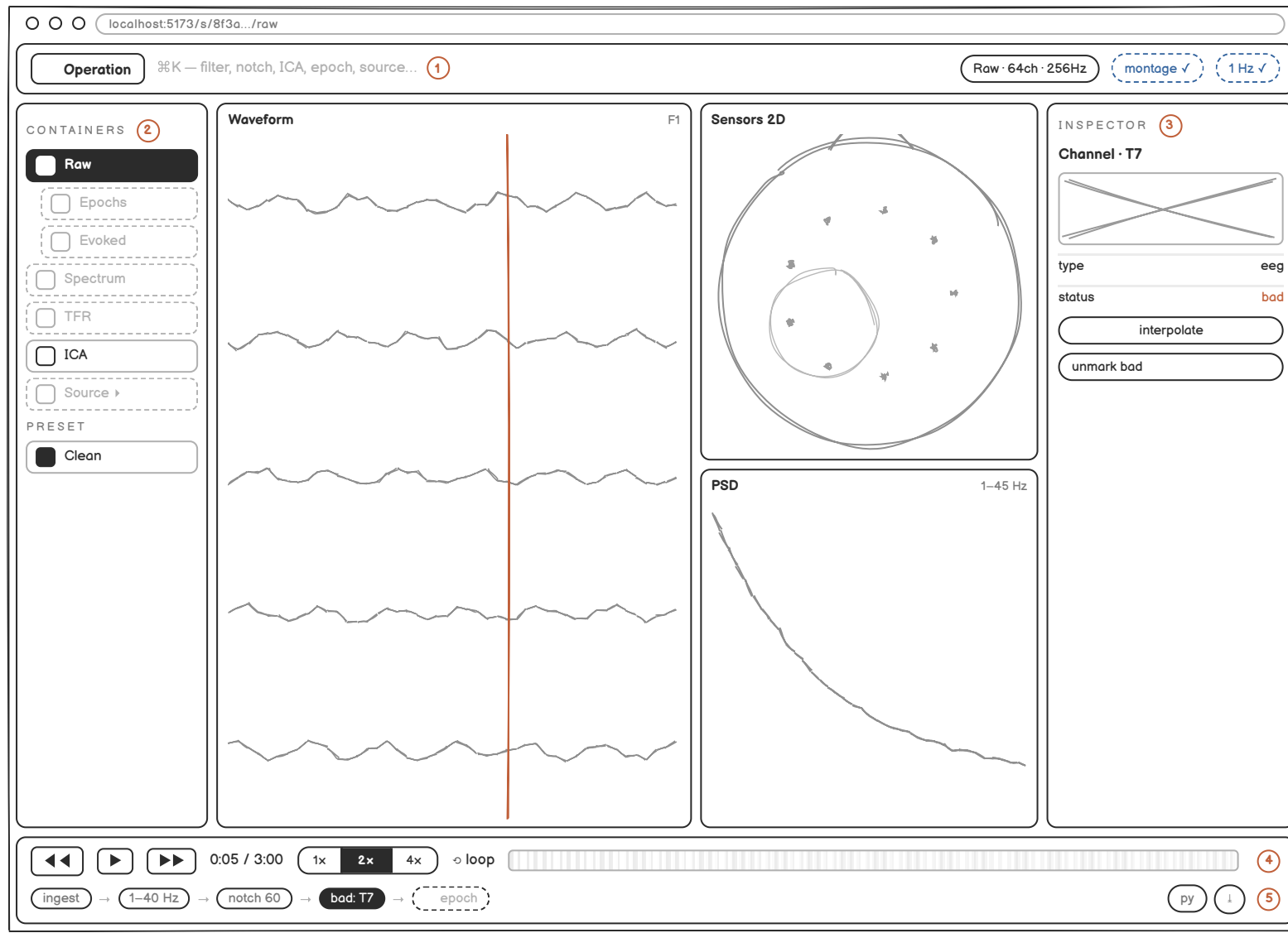
NOTES & DECISIONS

v1: auto-loads a synthetic 32-ch demo, upload is a secondary link, 7 formats. This proposes: a real entry point for every MNE input.

- 1 **Format + modality grid.** PROPOSED — reader registry (P0.10) covers ~25 formats; the Info decides which modality stages appear later. MEG/fNIRS/iEEG are real MNE inputs, not EEG-only.
- 2 **Sample datasets.** PROPOSED — mne.datasets.* one-click fetch (tracked job). Replaces the single hand-rolled synthetic demo; keep that as an offline fallback.
- 3 **Recent + bundle.** PROPOSED — sessions persist to ~/.eegvis/sessions/ (ledger + container snapshots), so refresh / reopen doesn't lose work. v1 loses everything on reload.
- 4 **Multi-file drop.** OPEN Q — group-by-stem on a folder drop vs an explicit "add sidecar" step for BrainVision/EEGLAB.
- 5 **Plugin path.** PROPOSED — eegvis.launch(raw) posts an in-memory Raw; every dev iteration and the real use-case go through the same door.

PRIOR ART

MNE's own raw.plot() is a matplotlib window per object. EEGLAB / Brainstorm are desktop GUIs tied to their own file stores. The bet: a browser session you can hand a URL to, seeded from any MNE reader.



DESKTOP ≥ 1200PX · SHELL/WORKSPACE.TSX — REPLACES TERMINAL.TSX FIXED GRID + TOOLBAR.TSX

NOTES & DECISIONS

v1: a fixed 5-panel grid, a preprocessing-only dropdown toolbar, a 9-item F-key strip, an 8-stat ticker that clips below ~950px. This proposes one coherent shell driven by registries.

- ① **Command bar.** [PROPOSED](#) — visible Operation button *and* ⌘K, both dispatch the operation registry (P0.2). Only ops whose input type matches the active container show. Capability pills replace the cramped ticker.
- ② **Container rail.** [PROPOSED](#) — the lineage DAG (P0.1). Click a node = active container; the canvas + offered ops reshape. Ghost = not created yet, click to see how. This is the whole nav model.
- ③ **Inspector.** [PROPOSED](#) — one context panel (channel / component / label / step / edge) replacing v1's scattered in-panel detail panes.
- ④ **Transport.** [OPEN Q](#) — real play/scrub/speed/loop driving the shared cursor. v1 has a Space-bar rAF sweep that pauses in background tabs and doesn't animate the spectral panels. Windowed PSD (P5) makes them move.
- ⑤ **Provenance strip.** [PROPOSED](#) — the DAG as a filmstrip (P0.6); click a node to time-travel, branch to compare. Export menu lives here.

PRIOR ART

Bloomberg / Persyst terminals: dense, keyboard-first, but fixed layouts. Observable / Jupyter: modular but no domain model. The bet: a domain-shaped shell (MNE's graph) with a modular canvas inside it.

localhost:5173/s/8f3a.../raw

Operation — input: Raw ❶

type to filter... "filt" — 3 matches

INGEST & ASSEMBLY

Crop · Resample · Concatenate · Anonymize

MONTAGE & CHANNELS

Set montagestd / file / digitized

Re-referenceavg / REST / bipolar

Set channel types · rename · pick / drop

Interpolate bad channels

⌵ closes loop

FILTER & REPAIR

Band-pass filterraw.filter

Notch filterfixed / spectrum_fit

Current source density (Laplacian)

Hilbert envelope · SSS / Maxwellmeg

ARTIFACT ID

Annotate muscle / amplitude / break

Detect EOG / ECG events · regress artifact

DECOMPOSE · SEGMENT · SPECTRAL · SOURCE

Fit ICA · SSP · Create epochs · PSD / TFR · Forward / Inverse ...

Band-pass filter

auto-form from schema ❷

l_freq (Hz)

1.0

h_freq (Hz)

40.0

method

FIR IIR

advanced

transition bw

auto

phase

zero

window

hamming

filter-response preview — magnitude + group delay ❸

Apply

Apply to a branch

Cancel

→ ledger + pipeline.py: raw.filter(l_freq=1.0, h_freq=40.0) ❹

MODAL OVER THE WORKSPACE · SHELL/COMMANDBAR.TSX + PARAMFORM.TSX

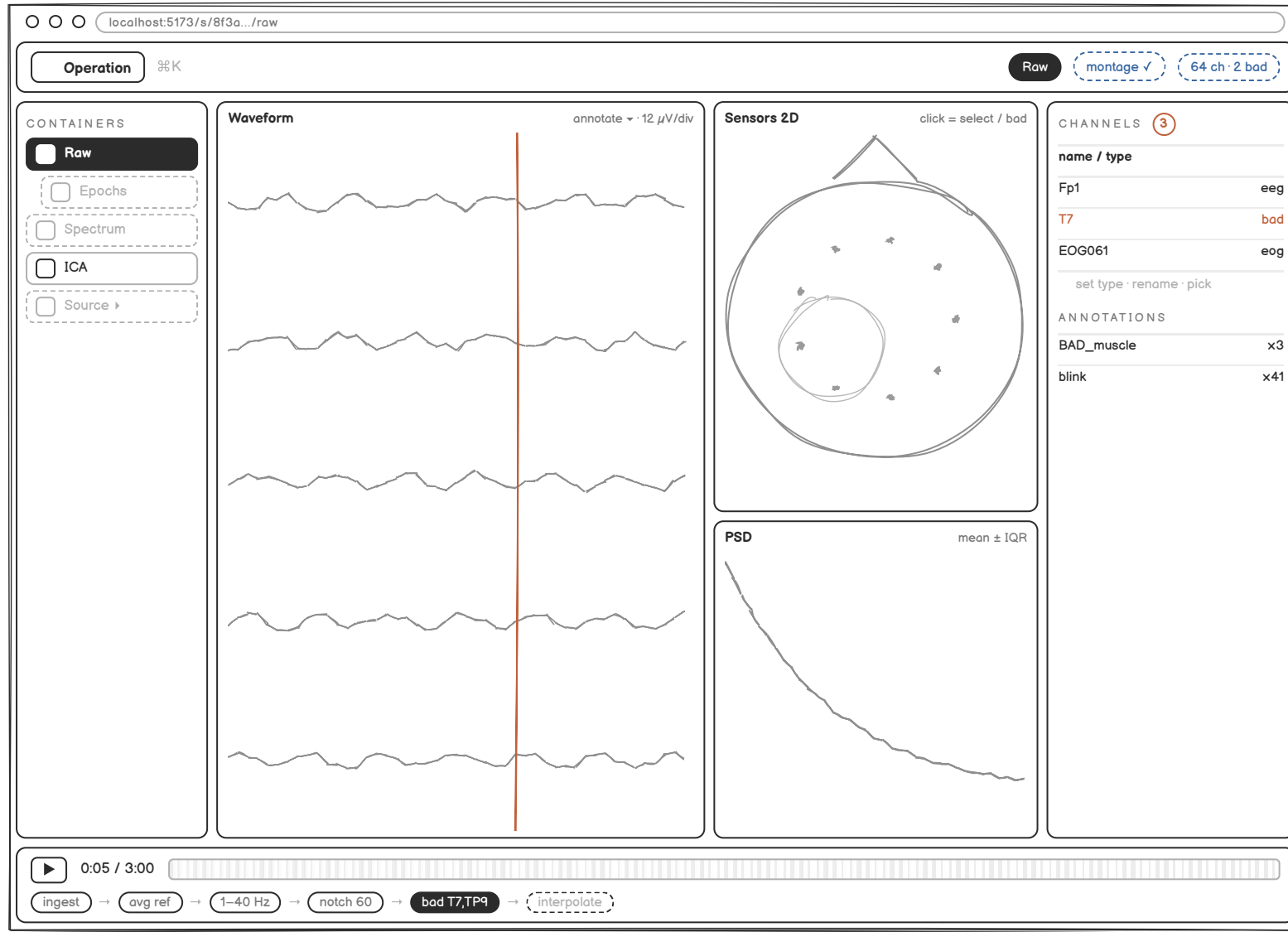
NOTES & DECISIONS

v1: preprocessing is 6 fixed toolbar dropdowns; everything else (ICA, epochs, export) is elsewhere or absent. A 36K palette was built then deleted. This proposes one surface generated from the registry.

- ❶ **Ops filtered by input type.** PROPOSED — the registry (P0.2) knows each op's input container. Active = Raw → you see Raw ops, grouped by MNE stage. Select Epochs → the list becomes reject / metadata / average.
- ❷ **Auto-generated form.** PROPOSED — <ParamForm> renders from the op's Pydantic → JSON schema (P0.8/P0.9). ~60 ops, zero bespoke form code. x-widget hints for channel-picker / band-picker.
- ❸ **Preview where cheap.** OPEN Q — filter response, ICA overlay, epoch count: which ops get a live preview vs apply-then-inspect. Default: preview if it's a pure function of params.
- ❹ **Apply → ledger + optional branch.** PROPOSED — every op appends a DAG node and its pipeline.py line. "Apply to a branch" forks (P0.6) to compare alternatives without losing the current line.
- ❺ **Long ops.** PROPOSED — ICA / forward / TFR / stats return a job_id; the form closes, a progress chip appears in the transport (P0.3).

PRIOR ART

Blender / Nuke node menus; VS Code command palette. The bet: the palette *is* the API surface — anything MNE can do to the active object is one fuzzy search away, and it writes the script.



THE "CLEAN" PRESET · CARDS FROM THE REGISTRY (P0.4)

NOTES & DECISIONS

v1: waveform + a hand-rolled "scalp field" + PSD + band heatmap + history, fixed. Mark-bad works but there's no interpolate_bads, no annotation editing, no channels view. **This proposes** the full Raw stage.

- 1 Waveform card.** **NOW** canvas scroller + shared cursor. **PROPOSED** annotation mode (draw / label / delete BAD_* spans), a real channel rail, WebGL renderer for 256-ch (P0.4).
- 2 Sensors 2D.** **PROPOSED** — plot_sensors idiom: named electrodes, click to select everywhere / toggle bad. Replaces the lumpy oblique "3D" blob, which was never a brain (see p.10).
- 3 Channels + annotations inspector.** **PROPOSED** — set type / rename / reorder / pick-drop; annotation counts. interpolate_bads is the **first** P2 op — marking a channel bad currently goes nowhere.
- 4 Artifact-ID ops.** **PROPOSED** — annotate_muscle_zscore, annotate_amplitude, EOG/ECG detection, regress_artifact, find_bad_channels_lof. All write Annotations or the bad list; all in pipeline.py.
- 5 Modality-aware.** **OPEN Q** — for MEG show a helmet layout + SSS stage; for fNIRS the optical-density chain. Driven by Info (P0.10). How much to specialise vs keep generic.

PRIOR ART

mne.Raw.plot() — the reference scroller (scaling, butterfly, annotation mode, filter-on-the-fly). This card should reach parity, in the browser, cursor-linked to every other card.

localhost:5173/s/8f3a.../ica

Operation ⌘K · input: ICA

picard · n=20 ICLabel ✓

CONTAINERS

☐ Raw

☒ ICA

ACTIONS

☐ auto-label

☐ find_bads_eog

☒ Apply -3

Components 20 · click inspect · dbl = exclude

4 more · scroll

Properties · IC 2 plot_properties

Clean vs raw overlay before apply

COMPONENT · IC 2

ICLabel eye 0.97

EOG corr 0.88

var. explained 11%

mark for removal

1-40 Hz → fit ICA picard → ICLabel → apply -[0,2,7]

NOTES & DECISIONS

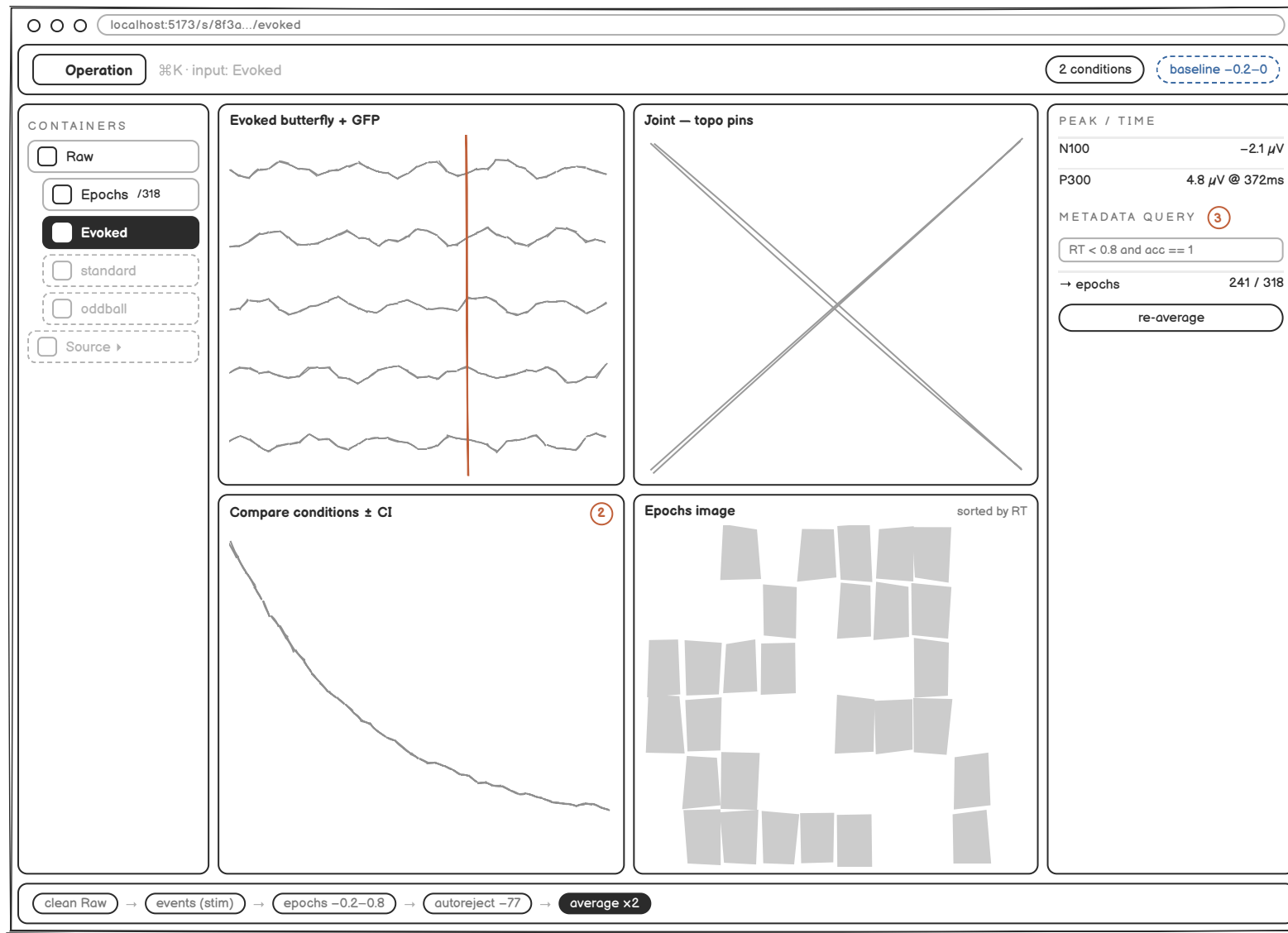
v1: fit → topo grid + variance → click for time-course + PSD → double-click to exclude → apply (reversible). No labels, no properties panel, no overlay, no confirm dialog. **This proposes** the full ICA loop.

- 1 **ICLabel.** [PROPOSED](#) — mne-iclabel classifies each component (brain / eye / muscle / heart / line / channel) with a probability. "Auto-exclude eye+muscle ≥ 0.8" as one action. The single biggest ICA time-saver.
- 2 **Properties card.** [PROPOSED](#) — the real plot_properties (topo + epochs-image + ERP + PSD + variance), server-rendered (P0.5). v1 shows only a time-course and a spectrum.
- 3 **Clean/raw overlay.** [PROPOSED](#) — a 2-leaf DAG compare (P0.6): ghost the replay(step-1) signal behind current so you see what apply removes *before* committing.
- 4 **Methods + find_bads_*** [PROPOSED](#) — picard / infomax; find_bads_eog/ecg/muscle + a plot_scores card. Fit on Epochs too.
- 5 **Apply guard.** [OPEN Q](#) — dialog vs the DAG making it cheap to undo. Default: a one-line confirm, since apply rewrites the signal (records a snapshot step already).

PRIOR ART

EEGLAB ICLabel + pop_selectcomps; MNE's ica.plot_components + plot_properties. The bet: the same workflow, but the exclusion set is a reversible ledger step and the overlay is live.

GATED: NEEDS MONTAGE + ≥1 HZ HIGH-PASS (P0.7)



EPOCHS + EVOKED ARE FIRST-CLASS RAIL NODES; CONDITIONS NEST

NOTES & DECISIONS

v1: epoching endpoints exist (from-annotations, fixed-length) and store an mne.Epochs — but **nothing in the UI creates or shows epochs**. Evoked doesn't exist at all. This is a dead limb.

- 1 **Events → Epochs → Evoked as containers.** [PROPOSED](#) — find_events / events_from_annotations / make_fixed_length_events; an editable events table + raster; Epochs with reject/flat/reject_by_annotation + autoreject.
- 2 **Evoked cards.** [PROPOSED](#) — butterfly + GFP, plot_joint, topo sequence / animate_topomap, plot_white, plot_compare_evokeds ($\pm$ CI). Server-rendered (P0.5), cursor-linked.
- 3 **Metadata.** [PROPOSED](#) — make_metadata + a pandas-query box → subset epochs → re-average without re-epoching. The reason to use Epochs at all.
- 4 **Field map.** [PROPOSED](#) — make_field_map + evoked.plot_field: the *principled* scalp/helmet field (spherical-spline), i.e. what v1's ScalpField3D fakes with inverse-distance weighting. No inverse pipeline needed.
- 5 **Averaging method.** [OPEN Q](#) — mean vs median vs robust default; how to present combine_evoked / grand_average for the single-subject tool.

PRIOR ART

ERPLAB, FieldTrip's ft_timelockanalysis. The bet: epoching is an *operation* in the same registry, the conditions are rail nodes, and the ERP is one branch of the same provenance graph as the cleaning.



CONTINUOUS SPECTROGRAM ON RAW USES THE SAME CARD

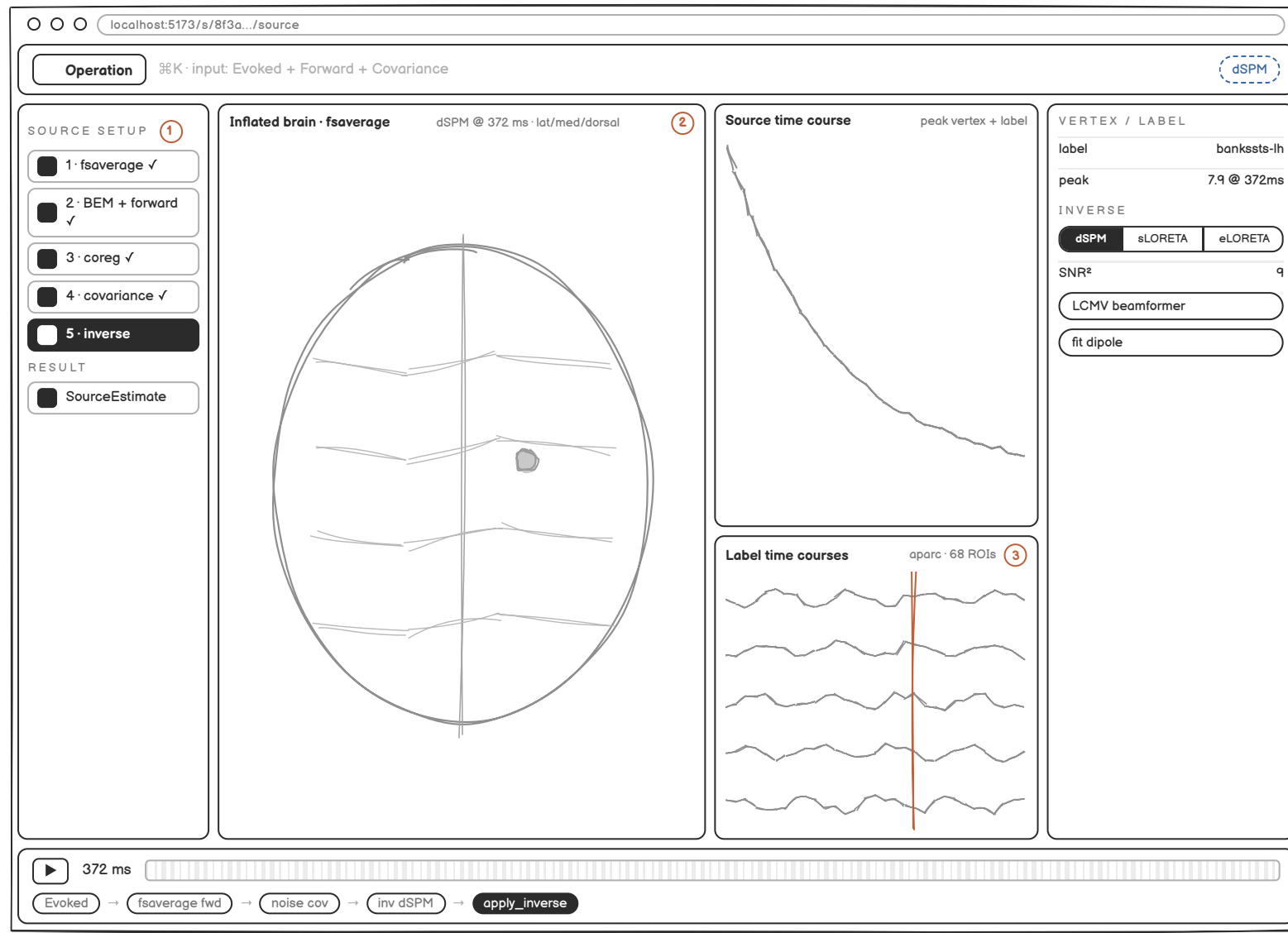
NOTES & DECISIONS

v1: Welch PSD (whole recording) + a band-power heatmap. No time-frequency at all — the single most-requested EEG view that's missing.

- 1 **TFR container.** [PROPOSED](#) — `tfr_morlet` / `tfr_multitaper` / `tfr_stockwell` on Epochs (or Raw for a continuous spectrogram). Method is a rail sub-choice; params via the auto-form.
- 2 **ITC.** [PROPOSED](#) — inter-trial coherence alongside power (both come from the same call). Its own card.
- 3 **Time-freq ROI → downstream.** [PROPOSED](#) — drag a box on the TFR → feed a topomap at that band/latency, or a cluster-stats contrast (P7). The inspector carries the ROI.
- 4 **Windowed PSD.** [PROPOSED](#) — add start/dur to the PSD op so Spectrum + Band-power animate as the transport sweeps. v1's spectral panels are static during playback.
- 5 **Baseline mode.** [OPEN Q](#) — `logratio` / `percent` / `zscore` default; where baseline is set (op param vs a card control).

PRIOR ART

EEGLAB `newtimef`, FieldTrip `ft_freqanalysis`. The bet: TFR is just another container off Epochs, on the shared canvas, feeding the shared stats and source stages.



GATED WORKSPACE · SERVER-RENDERED BRAIN FRAMES (P0.5 / P6) · REPLACES THE FAKE "SCALP FIELD"

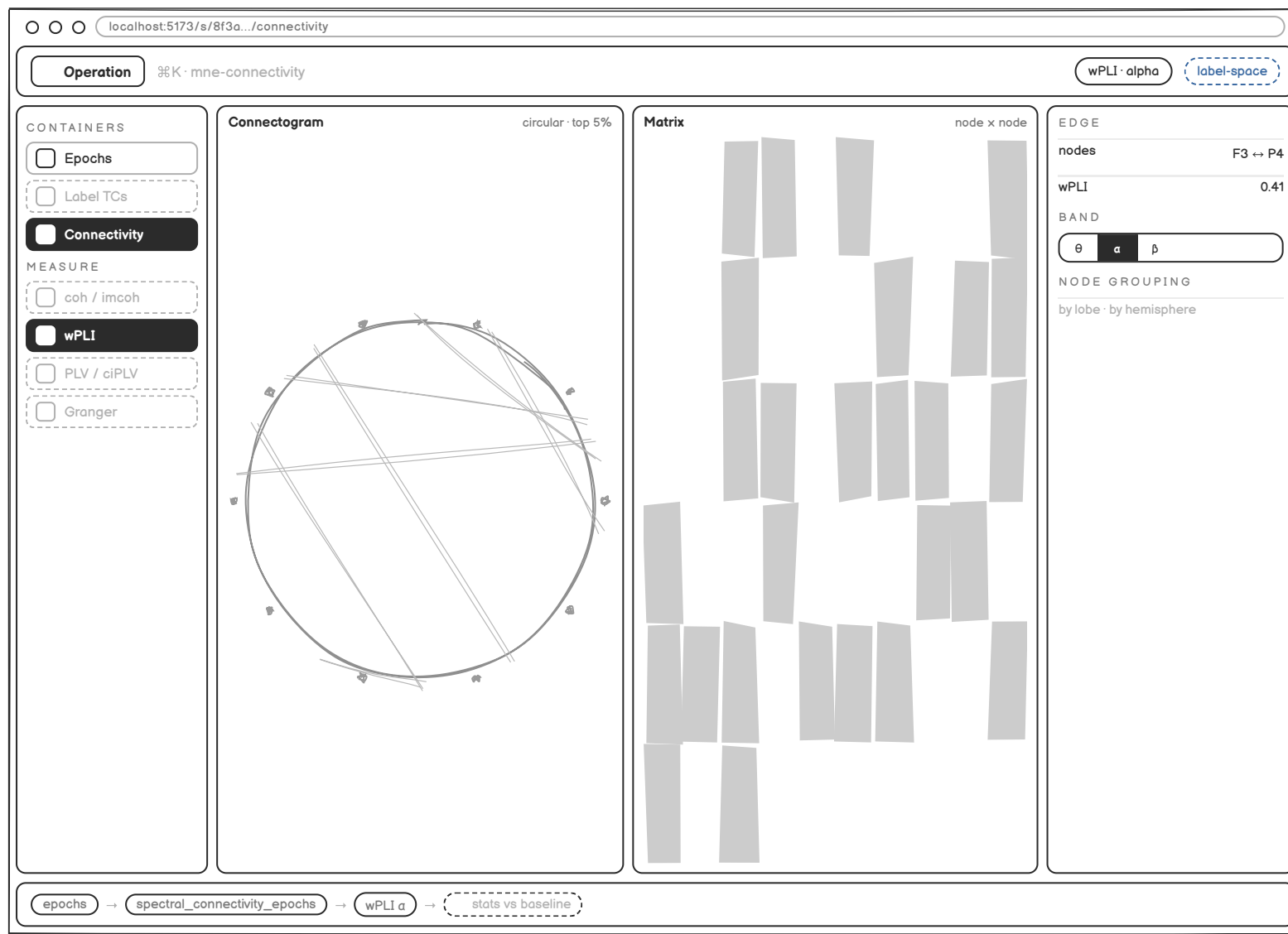
NOTES & DECISIONS

v1: the "Scalp field" panel is an inverse-distance interpolation of sensor voltages drawn as an oblique wireframe — **not a brain**, and there's no inverse pipeline behind it. This proposes the real thing, honestly gated.

- 1 Setup wizard. (PROPOSED)** — 5 gated steps (P0.7): fetch_fsaverage (~1 GB job) → BEM + make_forward_solution → coregistration → compute_covariance → make_inverse_operator. Each cached; each a ledger node.
- 2 The brain card. (PROPOSED)** — `stc.plot()` on an inflated fsaverage, PyVista rendered **offscreen server-side** to frames keyed to cursor time (like the ICA topo PNGs). No WebGL brain mesh (dies in a hidden tab — v1 lesson). Colourbar, hemi/view controls.
- 3 Label time courses. (PROPOSED)** — `extract_label_time_course` over a parcellation (aparc / a2009s / HCPMMP1 / Yeo). Click a label on the brain → its waveform; click a peak → the vertex.
- 4 Method + beamformers. (PROPOSED)** — MNE / dSPM / sLORETA / eLORETA as an inspector switch (recompute is cheap once the operator exists); LCMV / DICS and `fit_dipole` as alternative estimate ops.
- 5 Coreg without individual MRI. (OPEN Q)** — template fiducial fit for standard montages vs requiring digitized points; how loud the accuracy caveat needs to be. EEG + template head is coarse — the UI must not oversell it.

PRIOR ART

The linked `stc_class` tutorial uses the *sample* dataset's pre-computed inverse — it hides every step above. Brainstorm's source pipeline is the gold standard; this is a gated, reproducible subset of it. Honest intermediate first: `evoked.plot_field` (P6.0).



SENSOR- OR SOURCE-LABEL SPACE · SAME CARDS

NOTES & DECISIONS

v1: nothing. This proposes the mne-connectivity companion package as a workspace.

- 1 **Connectivity container.** [PROPOSED](#) — spectral_connectivity_epochs / _time: coherence, imcoh, PLV, ciPLV, PLI, wPLI, PPC, Granger. From sensor Epochs or source label time-courses.
- 2 **Cards.** [PROPOSED](#) — connectogram (plot_connectivity_circle), node x node matrix, 3D sensor-lines. Node grouping by lobe / hemisphere / parcellation.
- 3 **Also.** [PROPOSED](#) — envelope_correlation, phase_slope_index, vector_auto_regression.
- 4 **Threshold + stats.** [OPEN Q](#) — top-N% vs absolute vs permutation-thresholded edges as the default display.
- 5 **Decoding shares this workspace slot.** [PROPOSED](#) — mne.decoding (CSP, sliding estimator, temporal generalization, ReceptiveField) as a sibling "analysis" workspace, same card model.

PRIOR ART

FieldTrip connectivity, Brainstorm connectivity. The bet: it rides the same containers — sensor Epochs or source labels — and the same stats stage.

localhost:5173/s/8f3a.../provenance

Pipeline — DAG

click a node = time-travel · branch = compare ①

```

graph LR
    ingest --> avg_ref[avg ref]
    avg_ref --> f140[1-40 Hz]
    f140 --> notch60[notch 60]
    notch60 --> bad_interp[bad+interp]
    bad_interp --> ICA_picard[ICA picard → -3]
    ICA_picard --> epochs[epochs → evoked x2]
    epochs --> source_dSPM[source dSPM]
    ICA_picard -- branch B — compare --> ICA_infomax[ICA infomax → -4]
    ICA_infomax --> epochs
  
```

Export ②

PROPOSED

SCRIPT	— path to this node	copy · i
pipeline.py		
DATA		
cleaned	raw.fif / epo.fif / ave.fif	↓
EDF · EEGLAB · BrainVision		↓
summary.csv	— band power + log	↓
REPORT & BIDS		
report.html	— mne.Report, every stage	build
BIDS derivative —	write_raw_bids	wizard

pipeline.py

```

raw = mne.io.read_raw_fif('sub-04_raw.fif', preload=True)
raw.set_eeg_reference('average')
raw.filter(l_freq=1.0, h_freq=40.0); raw.notch_filter(60)
raw.info['bads'] = ['T7', 'TP9']; raw.interpolate_bads()
ica = ICA(n_components=20, method='picard').fit(raw); ica.exclude = [0,2,7]; ica.apply(raw)
epochs = mne.Epochs(raw, events, {'standard':1, 'oddball':2}, [-0.2, 0.8])...
  
```

preview

live from the DAG path ③

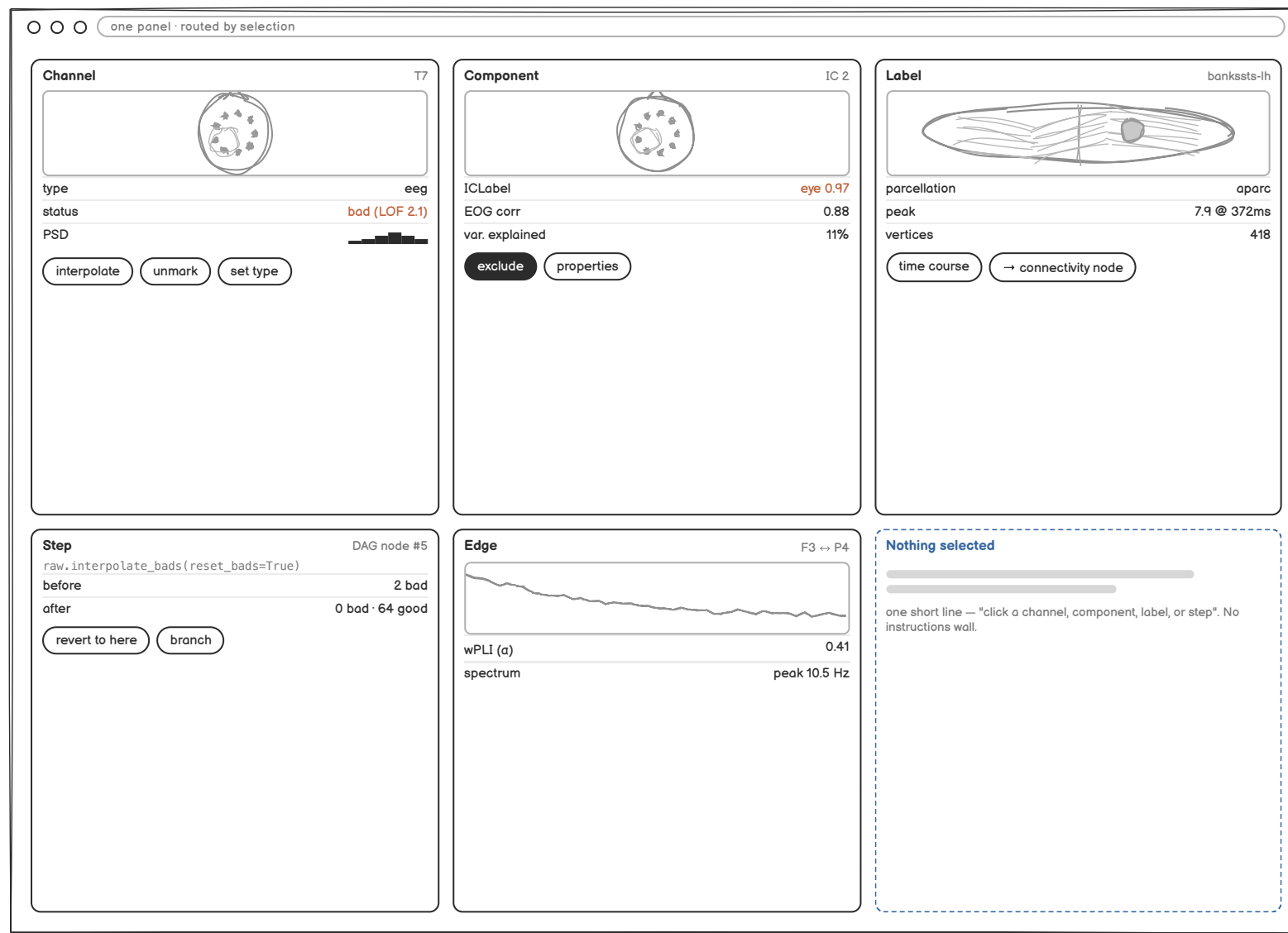
NOTES & DECISIONS

v1: a linear ledger (replay / revert / to_python), a History panel taking a permanent column, export = 3 <a download> links. Solid foundation. **This proposes** a branching graph + a real export surface.

- ① **Ledger → DAG.** **PROPOSED** — entries get a parent; branch (from_seq) starts an alternative line. Revert = check out a node without deleting siblings. The research loop ("try A, try B, keep B") stops destroying work.
- ② **Compare.** **PROPOSED** — pick two leaves → split card (filter A vs B on the waveform; two ICA solutions; two evokeds). Clean/raw overlay is the trivial case.
- ③ **pipeline.py per leaf.** **NOW** the codegen works; **PROPOSED** it emits the path to the selected node, and every registry op carries its own template (P0.2).
- ④ **Report + BIDS.** **PROPOSED** — mne.Report built from the DAG (add_raw / add_ica / add_evokeds / add_stc...); write_raw_bids + derivatives via a wizard.
- ⑤ **History panel leaves the grid.** **PROPOSED** — it's the bottom strip + this route now, not an always-on column (v1 audit P1).

PRIOR ART

Brainstorm's process history; nipy / MNE-BIDS-Pipeline for reproducible batch. The bet: the interactive session *is* the pipeline definition — no separate "now write the script" step.



ONE <INSPECTOR> · A ROUTER KEYED ON STORE.SELECTION

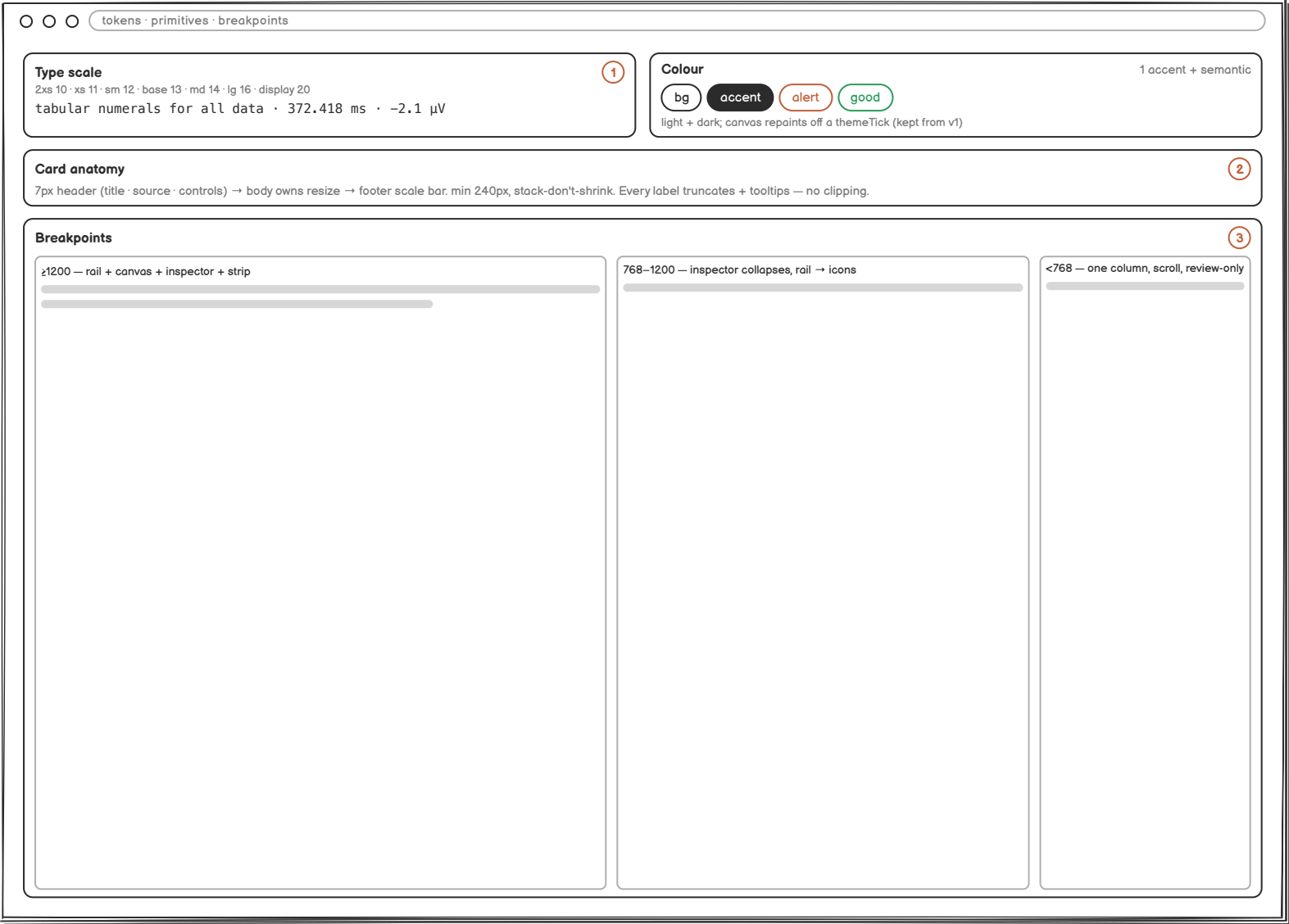
NOTES & DECISIONS

v1: detail views are scattered — the ICA panel has its own right-split, the waveform has a focus channel, nothing coordinates them. **This proposes** one panel, six states.

- 1 **One panel, routed by selection.** **PROPOSED** — `store.selection` is a tagged union (channel / component / label / step / edge / none). The inspector renders the matching sub-view. Every card writes to the same selection.
- 2 **Actions live here.** **PROPOSED** — `interpolate` / `exclude` / `revert` / "promote to connectivity node" are on the selected object, not in a far-away toolbar.
- 3 **Cross-links.** **PROPOSED** — select a label → "→ connectivity node"; select a channel → highlighted in every card. The selection is the glue between workspaces.
- 4 **Empty state.** **NOW** — one line, no instruction paragraph (v1 text-density lesson).
- 5 **Docking.** **OPEN Q** — fixed right rail vs a floating panel near the selection vs collapsible. Default: fixed right, collapsible < 1200px.

PRIOR ART

Figma's properties panel; browser devtools' element inspector. One consistent home for "what is this thing and what can I do to it".



NOTES & DECISIONS

v1 debt this fixes: "EEGvis" overlapping "32" in the ticker, "Bad channe" clipped, axis labels reading "125250375Hz", column headers overlapping, bottom tabs wrapping, no layout below ~1000px.

- 1 Every label owns its box.** **PROPOSED** — truncate + tooltip, never clip. Tabular numerals for all numeric data. One type scale, one spacing rhythm. This is P0.9 and it's non-negotiable.
- 2 Primitives as code.** **PROPOSED** — Card / Inspector / ParamForm / DataTable / Toolbar / Chip. "Cluttered and ugly" is a structural fix, done once, not per-panel polish.
- 3 Real breakpoints.** **PROPOSED** — 3 layouts by container width; the inspector and rail degrade gracefully; <768 is review-only (no overflow:hidden trap). v1 has none.
- 4 Keep from v1.** **NOW** — the sync-paint canvas substrate, themeTick repaint, Tailwind v4 tokens, the "dim don't blank on stale" rule. Those are good.
- 5 Aesthetic.** **OPEN Q** — "cool but clean": a restrained instrument look (thin rules, monospace data, one accent, generous negative space) vs more expressive. Wireframe stays neutral; pick in hi-fi.

PRIOR ART

Linear, Vercel dashboards, Ableton — dense, calm, one accent. Not Bloomberg's wall of orange. Information density from hierarchy and whitespace, not from filling every pixel.